TECTONICS, VOL. 16, NO. 3, PAGES 537-550, JUNE 1997

# Paleomagnetic evidence for rapid vertical-axis rotations during thrusting in an active collision zone, northeastern Papua New Guinea

Peter D. Weiler and Robert S. Coe

Department of Earth Sciences, University of California, Santa Cruz

**Abstract.** A paleomagnetic study of three thrust sheets of the fold and thrust belt north of the Ramu-Markham Fault Zone (RMFZ) indicates very rapid vertical-axis rotations, with differential declination anomalies related to tectonic transport of thrust units. Data from this investigation indicate depositional ages straddling the Brunhes-Matuyama reversal (780 ka) for the Leron Formation in Erap Valley. Net counterclockwise, vertical-axis rotations as great as 90° since 1 Ma have occurred locally in the Erap Valley area. These rotations appear to be kinematically related to shear across a tear fault within the foreland fold and thrust belt of the colliding Finisterre Arc, which in turn is aligned with and may be structurally controlled by a major fault in the lower plate. These data indicate that vertical-axis rotations occurred during thrusting; consequently, the actual rotation rate is likely several times higher than the calculated minimum rate. Such very rapid rotations during thrust sheet emplacement may be more common in fold and thrust belts than is presently recognized. Anisotropy of magnetic susceptibility data yields foliated fabrics with subordinate, well-grouped lineations that differ markedly in azimuth in the three thrust sheets. The susceptibility lineations are rendered parallel by the same bedding-perpendicular rotations used to restore the paleomagnetic remanence to N-S thus independently confirming the rapid rotations. The restored lineations are perpendicular to the direction of tectonic transport, and the minimum susceptibility axes are streaked perpendicular to the lineation. We interpret these anisotropy of magnetic susceptibility data as primary sedimentary fabrics modified by weak strain accompanying foreland thrusting.

## Introduction

Northern Papua New Guinea (PNG) is the site of an active, oblique collision between the Finisterre Arc terrane, an extinct volcanic arc lying in the Bismarck forearc, and the Australian continent. The collision has propagated from NW to SE through time and has uplifted the Finisterre terrane to form the Finisterre Mountains (Figure 1).

Copyright 1997 by the American Geophysical Union.

Paper number 97TC00493.
0278-7407/97/97TC-00493$12.00

This paper presents the results of paleomagnetic sampling of Quaternary, fluvial sediments of the Leron Formation, which record the syn collisional development of a foreland fold and thrust belt. The results indicate extremely rapid vertical-axis rotations of thrust sheets and provide information on the age of deposition and the development of magnetic fabrics in the collision suture.

Geologists now recognize vertical-axis block rotations, resolved through paleomagnetic declination anomalies, as a common consequence of crustal deformation, particularly in settings with strike-slip motion but also in settings characterized by thrusting [e.g., *Beck*, 1976; *Simpson and Cox*, 1977; *Schwartz and Van der Voo*, 1984; *Luyendyk et al.*, 1985; *Coe et al.*, 1985; *Wells and Heller*, 1988; *Fuller, et al.*, 1989]. Vertical-axis rotations have been shown to be disseminated across length scales up to hundreds of kilometers and are typically clockwise in dextral shear zones and counterclockwise in sinistral shear zones. Many paleomagnetic studies have assigned minimum rates for such rotations in degrees per million years, but we rarely know the true rate since a rock with a given declination anomaly may have rotated during a much smaller time interval than that bracketed by the paleomagnetic data. Published rotation rates from neotectonically active localities are generally much lower than rates documented in this study, implying that such rates may be much too conservative for rotations occurring during tectonic transport. Examples of such minimum Neogene rotation rates include 6.3°/m.y. and 5°-10°/m.y. in the Transverse Ranges of California [*Jackson and Molnar*, 1990; *Luyendyk*, 1991] and 10°/m.y. in the coastal range of eastern Taiwan [*Lee et al.*, 1991].

## Tectonic Setting

The orogenic belt in PNG is the result of a history of collisions along the northern edge of the Australian craton, with beds of the deformed Australian margin making up the southern part of the orogen and the northernmost part incorporating many accreted island arc terranes. The orogenic belt developed through the docking of a number of allochthons since the late Oligocene. *Pigram and Davies* [1987] have identified and interpreted 32 tectonostratigraphic terranes of New Guinea, each recording differing histories and depositional environments. At present, the Finisterre Arc terrane is colliding with the continent. The collision has propagated from NW to SE, and the present point of incipient collision is located at 149°30'E in the western Solomon Sea (Figure 1). The Ramu and Markham River valleys separate the rocks of continental affinity from the accreting Finisterre

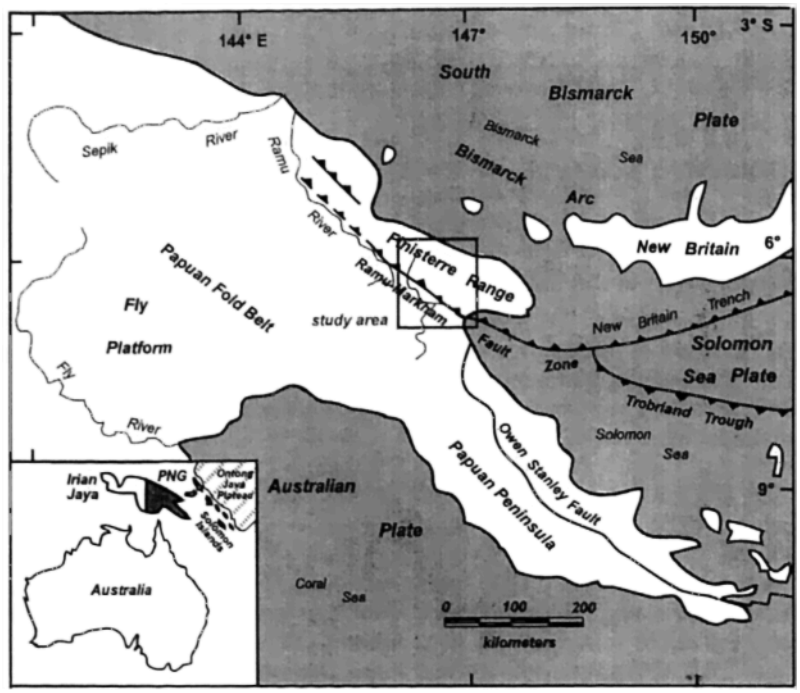



**Figure 1.** Regional tectonic setting of Papua New Guinea (adapted from *Abbott* [1993]).

Arc. On the basis of temporal shifts in sandstone provenance, *Abbott* [1993] reported the age of initial Finisterre terrane collision as between 3 and 3.7 million years.

## Geological Background

Among several clastic units deposited on the southern flank of the Finisterre Mountains, the youngest is the Quaternary Leron Formation [*Silver et al.*, 1991; *Liu*, 1993]. It includes fluvial, alluvial, lacustrine and shallow marine sandstones, mudstones and conglomerates [*Silver et al.*, 1991; *Liu and Crook*, 1993; *Liu*, 1993] as a fan delta complex [*Liu*, 1993]. The terrestrial depositional environment of the Leron Formation indicates that substantial uplift occurred between deposition of the bathyal Nariawang unit and the Leron Formation. Thus deposition of the Leron Formation post dates the onset of collision [*Silver et al.*, 1991; *Liu and Crook*, 1993].

Sandstones and conglomerates of the Leron Formation are volcanolithic, with their source in the Finisterre Mountains to the north [*Abbott et al.*, 1994]. The Leron Formation was deposited in the foreland of the collision, in a setting similar to the present-day Markham Valley, where compositionally similar alluvial material constitutes the Markham Formation [*Liu*, 1993]. Leron Formation sediments were folded and thrust in approximately 1000 m thick, north to northeast dipping sheets as the Finisterre thrust belt propagated southeastward. These thrust sheets are now exposed along the southern foothills of the Finisterre Range as part of an imbricate stack of steeply dipping thrust sheets that young from NE to SW [*Abbott et al.*, 1994] and do not exhibit the intense tectonic disruption of the older Erap complex rocks exposed to the north (Figure 2).

Owing to its terrestrial source, age determinations in the Leron Formation are sparse, and *Liu* [1993] has devoted considerable effort to exploring techniques to address this problem. The Leron Formation is devoid of age diagnostic fauna, and, apart from carbon-bearing matter and local volcanic ash layers, there are few materials suitable for Quaternary dating methods. It is presumably time-transgressive from NW to SE [*Silver et al.*, 1991; *Liu*, 1993] and is constrained by the youngest age from the underlying Nariawang unit which is around 1.1 to 1.3 Ma in this area [*Abbott and Silver*, 1991]. Therefore magnetostratigraphic dates would prove valuable to workers in this area and are an important result of this study.

For this study, samples were collected from thrust sheets exposed in the valley of the Erap River (Figure 2). Deformation has produced 50° to 60° N to NE dipping homoclines that are typical of the stack of thrust sheets of Leron Formation exposed in several river valleys along the southern foothills of the Finisterre Mountains. These faults are small-scale relative to large thrust faults displacing the older Erap complex to the north. The fault zones separating Leron Formation thrust sheets exposed in the Erap Valley are a few meters to a few tens of meters wide and do not contain the exotic blocks of older lithologies found in the several hundred meter wide mélange zones of the Erap Complex.

The region sampled for this study is near the intersection of at least two major faults. The Ramu-Markham Fault Zone (RMFZ) is a NW-SE oriented, north dipping thrust generally regarded as the plate boundary in this collision. The RMFZ is thought from earthquake evidence to have a pure thrust sense of motion [*Abers and McCaffrey*, 1994]. The left-lateral Wampit Fault [*Tingey and Grainger*, 1976] is nearly normal to the RMFZ, controlling the Wampit River Valley on

19449194, 1997, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/97TC00493 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

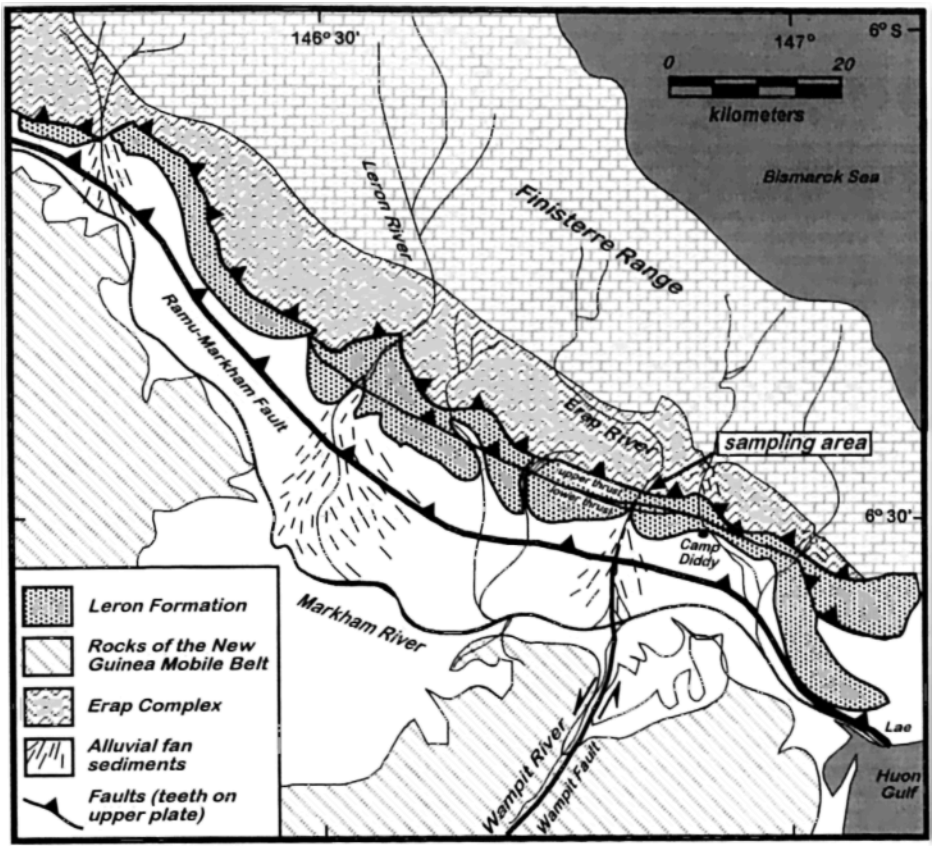



**Figure 2.** Generalized geologic map of study area showing sampled section in thrust sheets along the Erap River and the Wampit Fault controlling the Wampit River to the south.

the south (lower plate) side of the main thrust (Figure 2). North of the RMFZ is a lineament photo mapped by the Geologic Survey of PNG [*Tingey and Grainger*, 1976] through the Erap Valley which is collinear with the Wampit Fault. This fault may extend across the RMFZ, displacing units on the upper plate (R.H. Findlay, personal communication, 1995). Though beds of the Leron Formation can be correlated across the Erap Valley, displacement on a northern extension of the Wampit Fault, which is not exposed at the surface in this portion of the Erap Valley, is likely responsible for the vertical-axis rotations in the Erap Valley area.

## Sampling and Laboratory Procedures

Samples from Erap Valley were collected throughout a river exposure of three Leron Formation thrust sheets with a total stratigraphic thickness of 970 m (Figure 3). The lower (southern) thrust sheet has an exposed stratigraphic thickness of about 360 m. The central thrust is exposed over 86 m, and the upper (northern) thrust sheet is exposed over 524 m of section. The three thrust sheets are separated by several meters of pervasive layer-parallel shearing noted in outcrop.

We recognized large differential rotations in paleomagnetic measurements of hand samples collected for reconnaissance magnetostratigraphy and speculated that these thrust faults might be acting as rotation surfaces. We returned to Erap Valley and collected 2.5 cm diameter cores at approximately 10 m intervals with occasional gaps of several tens of meters where exposure was poor. We regularly verified magnetic compass directions throughout the sampling area by sun compass. We have analyzed 108 specimens through the section, which at estimated deposition rates for the Leron Formation of 6-10 m/ka [*Brierley et al.*, 1993] span ample time to average secular variation of the geomagnetic field.

We measured remanent magnetic directions and intensities with a 2G cryogenic magnetometer, applying both alternating field (AF) and thermal demagnetization techniques to a total of 117 Erap Valley samples and using principal component analysis to obtain best fitting magnetization directions. We did additional tests on six specimens to determine the magnetic mineralogy and domain state of the magnetic grains. hysteresis measurements using a Princeton Measurements Corporation alternating field gradient magnetometer and low-field thermomagnetic measurements to determine Curie points and degree of alteration of magnetic minerals during heating using a Kappabridge KLY-2 with CS-2 oven. We also measured anisotropy of magnetic susceptibility (AMS) for 101 samples using a Kappabridge KLY-2 susceptibility meter.

## Results

### Demagnetization Results

Representative Zijderveld vector plots of demagnetizations, including normalized magnetization ($J/J_o$) plotted versus

19449194, 1997, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/97TC00493 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

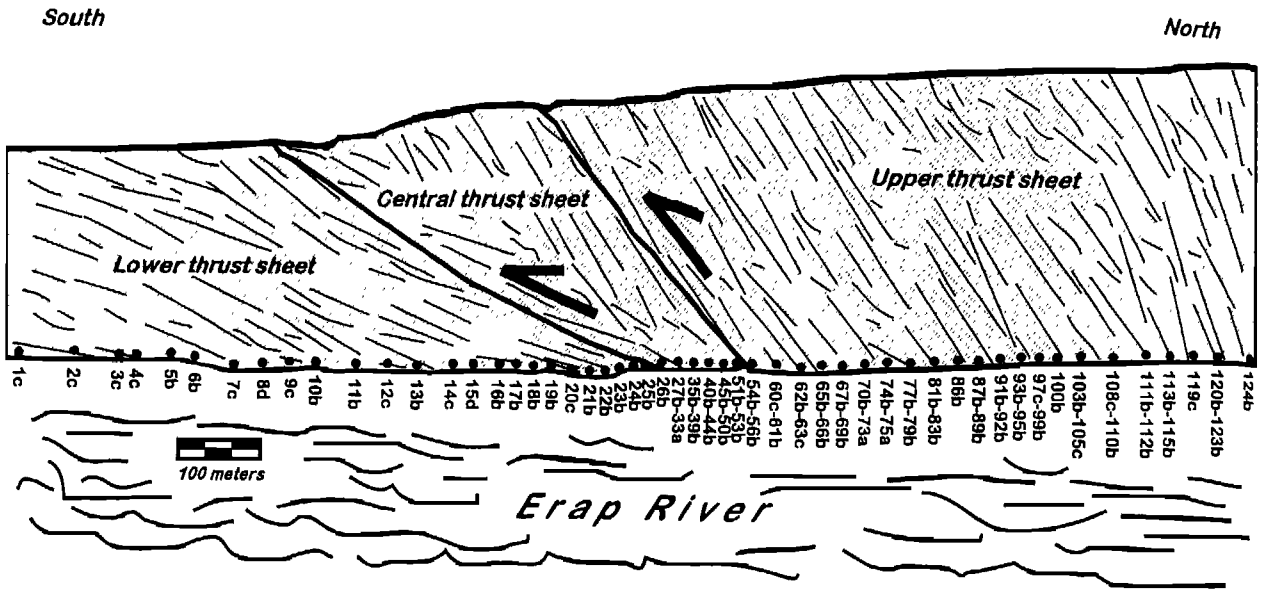



**Figure 3.** Summarized cross section illustrating Erap River exposure geometry. Fine dipping lines on thrust sheets indicate dip of bedding; bold lines indicate thrust faults with sense of motion. Sample locations are shown at base of outcrop along shoreline of the Erap River.

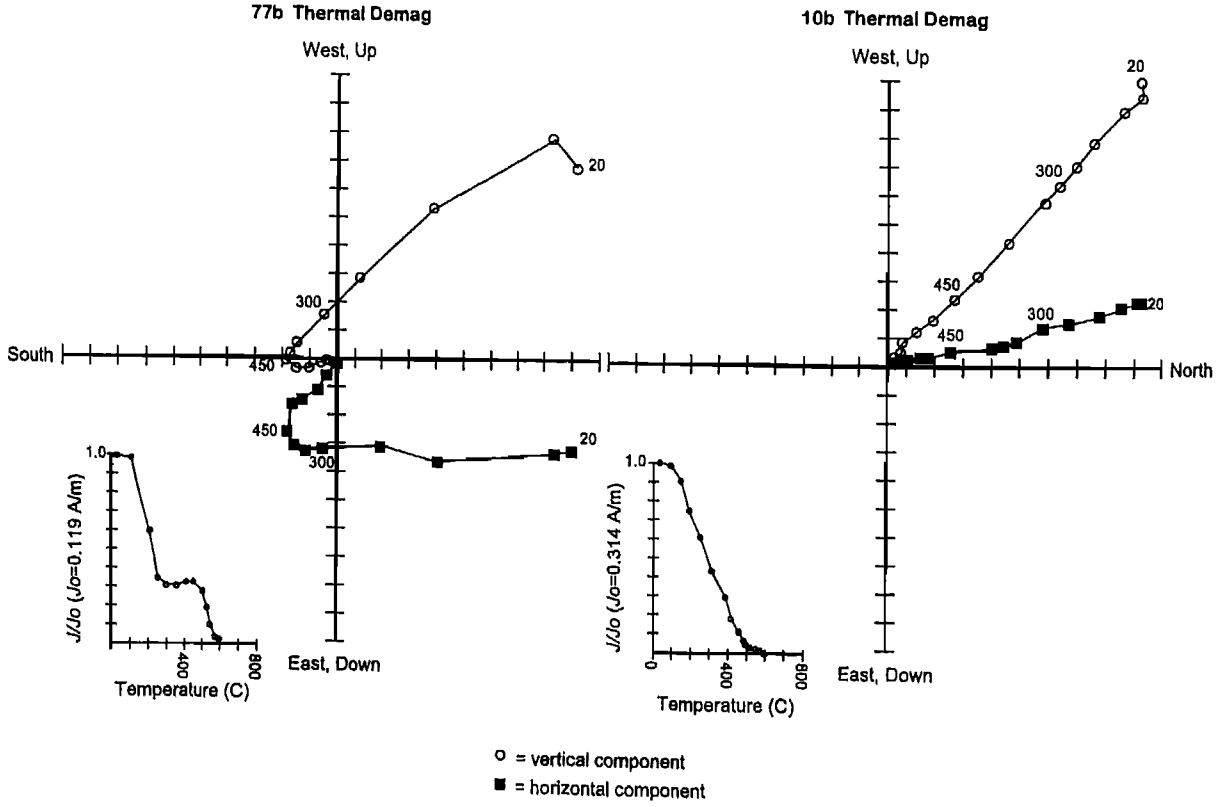



**Figure 4.** Representative vector plots of thermal demagnetizations of Erap Valley Leron Formation: (left) upper thrust and (right) lower thrust.

19449194, 1997, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/97TC00493 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

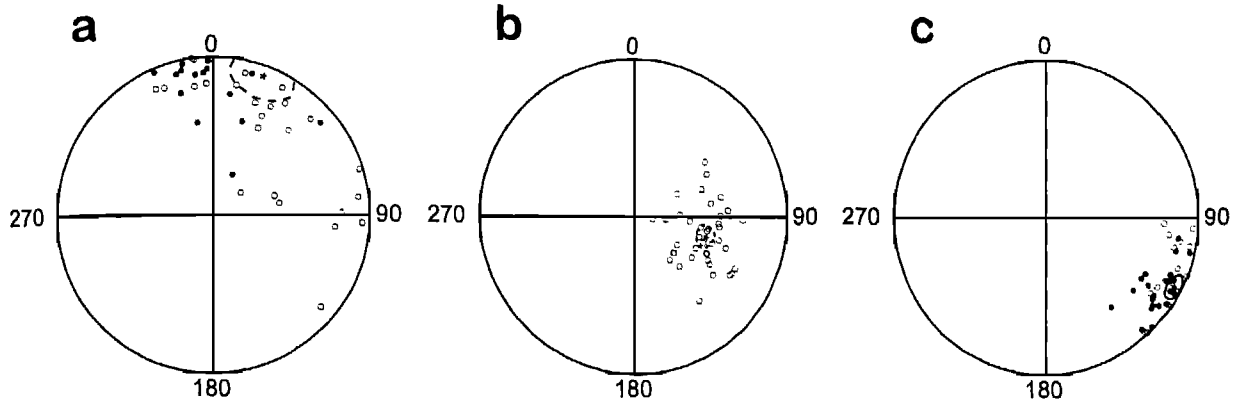



**Figure 5.** Equal area projections of characteristic remanent magnetization (ChRM) directions for upper Erap thrust: (a) natural remanent magnetization (NRM), (b) in situ, and (c) tilt-corrected directions. Open symbols are upper hemisphere projections. Stars indicate mean direction with surrounding circle of 95% confidence. Site statistics are given in Table 1.

demagnetization temperature, are presented in Figure 4, and equal area projections of natural remanent magnetization (NRM) directions and characteristic remanent magnetization (ChRM) directions for the three separate structural units are given in Figures 5 through 7.

Vector plots shown in Figure 4 illustrate that thermal demagnetization of 90% of the Erap Valley rocks provides consistent determination of ChRM directions, with secondary components cleanly removed and magnetization decaying reasonably linearly toward the origin. NRM intensities averaged of the order of $10^{-1}$ A/m, and demagnetizations were nearly complete by 600°C. The ChRM is usually isolated by 300°C and 400°C, after removal of a low-temperature component with a present geomagnetic field direction typical of viscous remanent magnetization (VRM). Nine additional specimens were demagnetized using AF techniques. Vector directions which are in general agreement with those obtained by thermal demagnetization can be identified, but AF demagnetization did not yield univectorial decay of the ChRM component, and consequently, AF results were not used in determination of site mean directions.

Ninety-six of the 108 thermally demagnetized specimens possess an identifiable stable ChRM. The demagnetization behavior of eight (chiefly coarse-grained) specimens was noisy, often yielding no vector segments, and the remaining four specimens were collected from within the sheared fault zones between thrust sheets, where ChRM directions were found to be erratic.

### Paleomagnetic Directions

ChRM directions from 39 specimens on the upper (northern) thrust sheet are shown in Figure 5 and in Table 1. The ChRM yields a mean declination (D) of 117.6° and inclination (I) of 5.9° in tilt-corrected (stratigraphic) coordinates, with a confidence limit ($\alpha_{95}$)=5.2° and dispersion estimate ($k$)=20.1. The tilt-corrected mean inclination is much closer to the axial dipole inclination (13°) at this latitude than the in situ (geographic) inclination (-51.7°). This suggests that samples from the northern thrust sheet acquired their ChRM prior to tilting. A conventional fold test is not possible since bedding is uniformly dipping to the north throughout the sampled section.

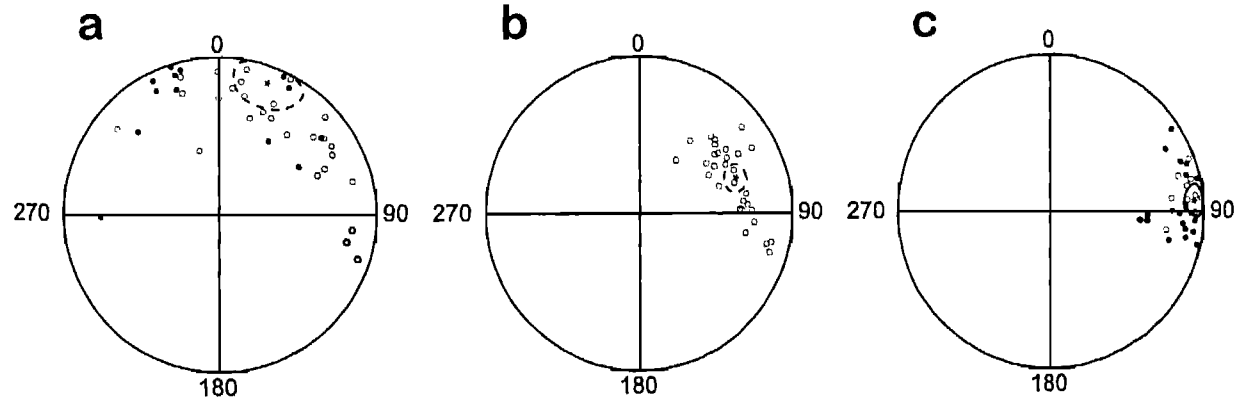



**Figure 6.** Equal area projections of ChRM directions for central Erap thrust: (a) NRM, (b) in situ, and (c) tilt-corrected directions (open symbols are upper hemisphere). Site statistics are given in Table 1.

19449194, 1997, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/97TC00493 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

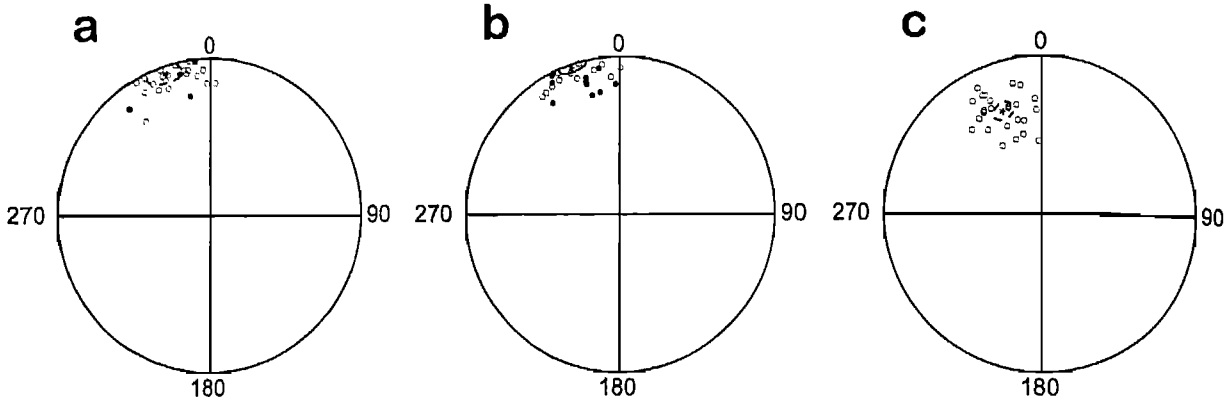



**Figure 7.** Equal area projections of ChRM directions for lower Erap thrust: (a) NRM, (b) in situ, and (c) tilt-corrected directions (open symbols are upper hemisphere). Site statistics are given in Table 1.

Thirty-one specimens from the central thrust give a ChRM mean of D=85.8° and I=6.3° in tilt-corrected coordinates with $\alpha_{95}$=6.5° and $k$=17.0. As in the upper thrust, the tilt-corrected mean inclination is about 6° shallower than the axial dipole field direction, whereas the in situ mean gives an unreasonably steep inclination for a dipolar field (-34.1°) (Figure 6). Shallow inclinations displayed by data from the upper and central thrust sheets may result from a combination of compaction and a slight southward depositional dip.

Twenty-six specimens from the lower (southern) thrust yield a mean direction in tilt-corrected coordinates of D=338.4° and I=-31.3°, with $\alpha_{95}$=5.0° and $k$=33.2. The mean declination close to north suggests normal polarity, but the mean inclination is 18° more steeply upward than the axial dipole inclination of -13°. This difference is highly significant and much too large to explain by northward translation of PNG in the last 1 m.y. or by our samples spanning too short a time to average secular variation of the geomagnetic field.

**Table 1.** ChRM Means And Site Statistics

| | N/n | Dec, deg | Inc, deg | $\alpha_{95}$, deg | $k$ | $R$ |
|---|---|---|---|---|---|---|
| | | | *Upper Erap Thrust* | | | |
| In situ | 39/41 | 105.4 | -51.7 | 5.4 | 19.1 | 37.01396 |
| Tilt-corrected | 39/41 | 117.6 | 5.9 | 5.2 | 20.1 | 37.10895 |
| Strike*: 215° - 244° | dip: 58° - 70° | | | | | |
| | | | *Central Erap Thrust* | | | |
| In situ | 31/40 | 72.1 | -34.1 | 6.7 | 16.0 | 29.12718 |
| Tilt-corrected | 31/40 | 85.8 | 6.3 | 6.5 | 17.0 | 29.23668 |
| Strike*: 215° - 238° | dip: 54° - 69° | | | | | |
| | | | *Lower Erap Thrust* | | | |
| In situ | 26/27 | 341.3 | 1.6 | 5.6 | 27.1 | 25.07589 |
| Tilt-corrected | 26/27 | 338.4 | -31.3 | 5.0 | 33.2 | 25.24641 |
| Strike*: 215° - 280° | dip: 20° - 43° | | | | | |
| | | | *Camp Diddy* | | | |
| In situ | 4/4 | 217.5 | -45.8 | 27.3 | 12.3 | 3.75667 |
| Tilt-corrected | 4/4 | 191.5 | -0.7 | 13.8 | 45.4 | 3.93396 |
| Strike*: 236° | dip: 64° | | | | | |

N/n, total number of samples/number of samples revealing characteristic remanence; Dec, mean declination; Inc, mean inclination; $\alpha_{95}$, half angle of the 95% confidence cone for mean direction; $k$, dispersion estimate for mean direction; and $R$, length of resultant vector.

* Strike is recorded to left of downdip direction.

19449194, 1997, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/97TC00493 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

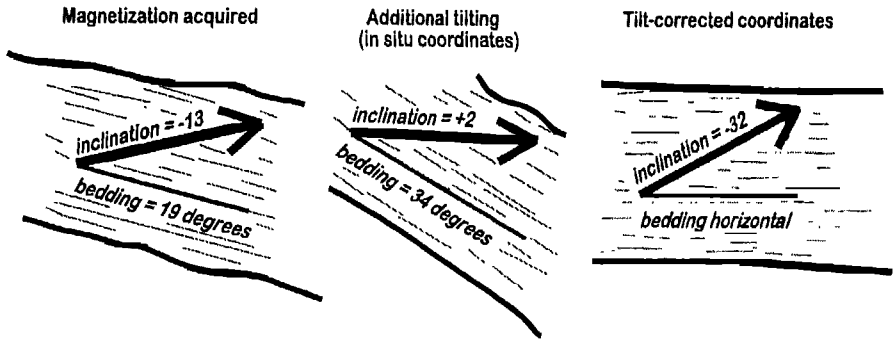



**Figure 8.** Sketch illustrating the effect of continued tilting of the lower thrust sheet after acquisition of a magnetization at an initial mean dip of 19° (see text).

It is of the wrong sense to explain by compaction shallowing. Moreover, the magnetization direction does not appear to be a posttilting overprint because the mean in situ inclination is nearly 15° more downward than the expected axial dipole inclination (Table 1).

To determine whether or not the peculiar mean inclination given by the lower thrust sheet is the result of an incompletely removed secondary component, we fit 27 great circles to the NRM vectors above 400°C using the statistical method of *McFadden and McElhinny* [1988]. The best fit intersection point using this analysis is D=337.5° and I=-32.3°, with $\alpha_{95}$=7.3° and $k$=15.8. This result is remarkably similar to the vector component direction and thus contradicts the hypothesis of an incompletely removed secondary component.

The lower thrust sheet data are well-explained by a syntilting magnetization or remagnetization in a normal polarity dipole field, if bedding had an initial mean northward dip of about 19° when magnetized in a field with the expected mean inclination of -13° (Figure 8). Further tilting to the average bedding dip for our samples of 34° yields the observed +2° in situ inclination. Restoration of bedding to the horizontal in this model gives a tilt-corrected inclination of -32°, within 1° of the actual value. In addition, lower thrust sheet ChRM directions are univectorial, decaying over 30% of the NRM toward the origin. This, coupled with the observation that these directions are not streaked, as might be expected for mixed primary and remagnetized directions, and the extremely close agreement between vector and great circle fits, indicates a true syn tilt magnetization.

Since the lower thrust sheet is the only structural unit with significant variation in bedding attitudes, we applied the McFadden Fold 2 test [*McFadden*, 1990] to the ChRM directions within this unit. The results yield a SCOS2 statistic of 10.1 in in situ coordinates, greater than the 99% critical value of 8.4. The minimum SCOS value is 0.01 at 46% unfolding (16° average dip), quite close to the dip assumed in the simplified syn tilting model illustrated in Figure 8. The relatively good agreement between the calculated syn folding and syn tilting dips indicates that the large-scale tilting and small-scale folding of the bedding probably occurred together, rather than sequentially.

Since the declination anomalies in the Erap Valley are so extreme and the expected dipolar inclination is quite shallow (about 13°) at this latitude, we also report results for four reconnaissance samples collected at Camp Diddy, about 7 km southeast of Erap Canyon (Figure 2 and Table 1). The Camp Diddy samples were collected in the upper Leron Formation section, stratigraphically within a few tens of meters of the thrust contact overlying the northernmost thrust sheet in Erap Canyon. As shown in Table 1, the results from Camp Diddy indicate unambiguous reversed polarity ChRM, with an 11° clockwise declination anomaly.

### Polarity Uncertainty and Age of the Leron Formation

ChRM directions from the upper and central thrusts in Erap Valley display only one polarity. As discussed above, the declination anomalies are so extreme and the inclinations are so shallow that it is not possible to determine polarity with certainty. The more reasonable hypothesis from magnetic data is that the upper and central thrusts show a reversed polarity ChRM, with mean declinations rotated counterclockwise 62° and 94°, respectively. This interpretation is consistent with the shallow positive inclinations for these data sets (only 7° different from the expected axial dipole value) and with four unambiguous reversed-polarity samples of the Leron Formation collected at Camp Diddy, 7 km to the SE (Table 1). The opposite hypothesis of normal polarity is less likely because it would entail ChRM inclinations that differ from the expected axial dipole inclinations by 20°. Moreover, although vector plots of demagnetizations do not indicate strong antiparallel secondary components, which would be expected if a normal polarity VRM was acquired after deposition but long before rotation, a high degree of scatter in NRM directions also suggests an underlying reversed polarity ChRM.

If the Leron upper and central thrusts at this locality are of reversed polarity, then the geomagnetic polarity timescale [e.g., *Cande and Kent*, 1992] implies that the Leron Formation sampled at these localities was deposited prior to 780 ka. As discussed above, paleomagnetic directions from the lower thrust sheet suggest a secondary normal polarity magnetization acquired during tilting associated with early thrusting. The lower thrust sheet is typical of the uppermost conglomeratic (proximal alluvial fan) facies of *Liu and Crook* [1993], so the depositional age of the Leron Formation in Erap Valley most likely straddles the Brunhes-Matuyama polarity reversal. Alternatively, it may be a little older than 780 ka.

19449194, 1997, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/97TC00493 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

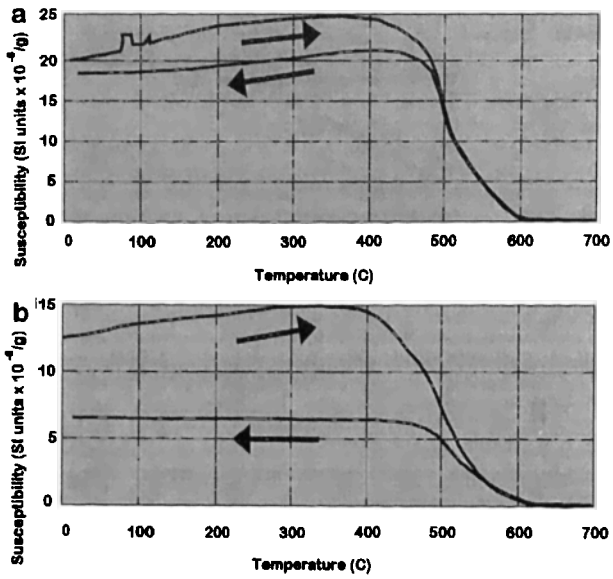



**Figure 9.** Thermomagnetic curves from (a) upper (specimen 108b) and (b) lower (specimen 9c) Erap Valley thrusts (noise in upper thrust heating curve at ~100°C is due to moisture in sample holder). Susceptibility is given in SI units times $10^{-6}$/g.

### Hysteresis and Thermomagnetic Behavior

To assess the possibility that the Erap thrust sheet results might be influenced by differing magnetic mineralogy, we obtained hysteresis curves for several specimens from each thrust sheet using an alternating gradient force magnetometer. The magnetic grains in all specimens analyzed saturated in applied fields of less than 300 mT, indicative of dominantly low-coercivity minerals such as magnetite. These rocks gave consistent ratios of saturation remanence to saturation magnetization ($J_r/J_s$) of about 0.12, typical of pseudosingle domain state. Thermomagnetic curves showing temperature dependence of low-field magnetic susceptibility (Figure 9) indicate Curie temperatures typical of magnetite (560°C-580°C), with a significant amount of a lower Curie point or metastable phase in the specimens from the lower thrust sheet. The sizeable decrease in magnetic susceptibility of lower thrust specimens following heating to 700°C suggests that this phase might be maghemite, which inverts to much less magnetic hematite on heating above 400°C.

### Anisotropy of Magnetic Susceptibility

We measured the anisotropy of magnetic susceptibility (AMS) tensor of undemagnetized specimens from 101 of the

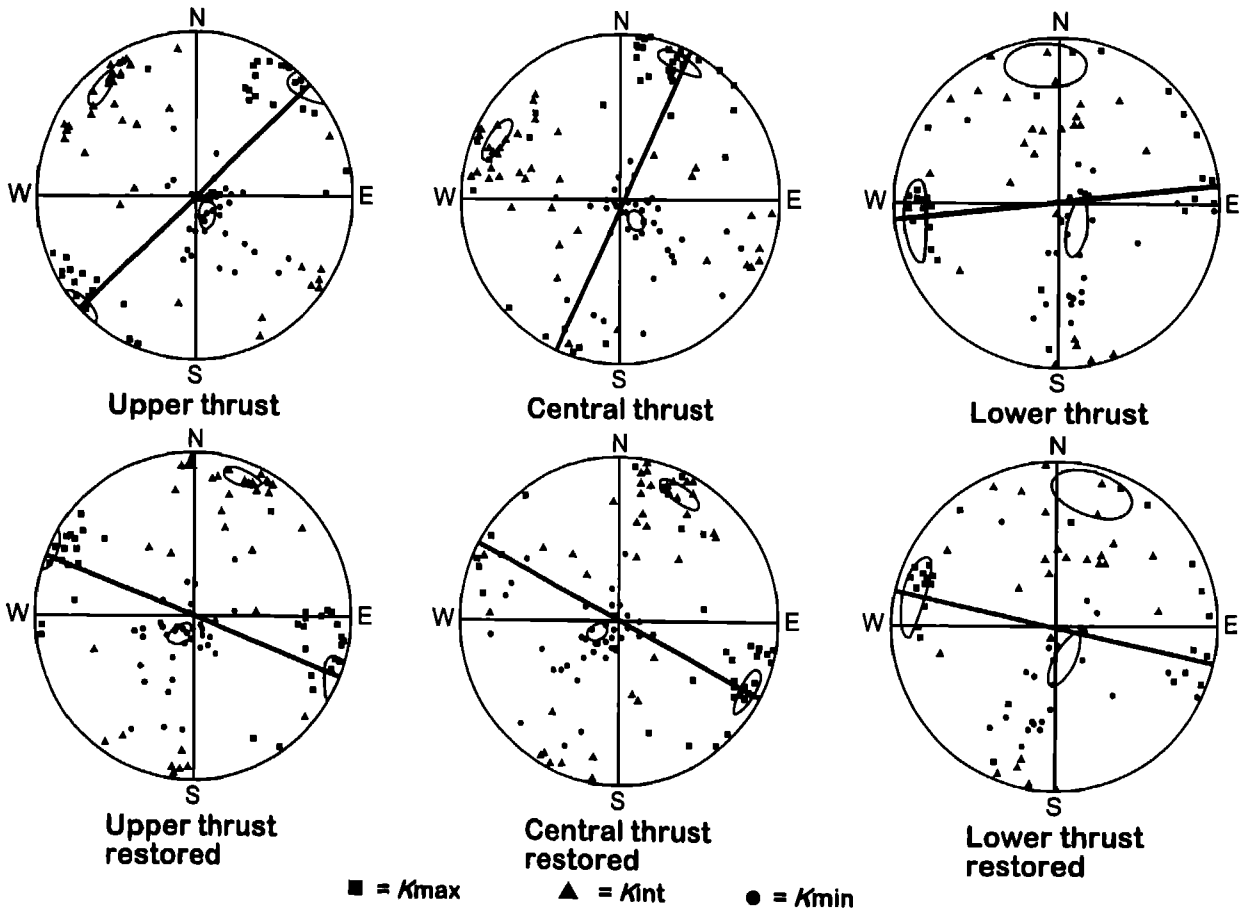



**Figure 10.** Equal area projections of anisotropy of magnetic susceptibility (AMS) principal axes for Erap Valley thrust units: (top plots) in situ and (bottom plots) restored using remanence declination anomalies. Bold lines indicate mean $K_{max}$ direction (magnetic lineation). Confidence ellipses from Hext/Jelinek statistical analysis (see text).

19449194, 1997, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/97TC00493 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

same cores used for studying the remanence. We also analyzed 36 specimens that had been partially thermally demagnetized to detect any enhancement of AMS fabrics [*Tarling and Hrouda*, 1993] but observed negligible change in AMS for the heated specimens. Magnetic fabrics, determined from the orientations of principal axes and the shapes of AMS ellipsoids have been used as proxy petrofabrics to discern mechanisms of sediment deposition [*Taira*, 1989] and post depositional deformation styles and geometries [*Kanamatsu et al.*, 1996; *Hrouda*, 1991]. We have characterized the magnetic fabrics by applying the following conventional parameters: lineation $L = K_1/K_2$ [*Balsey and Buddington*, 1960]; foliation $F = K_2/K_3$ [*Stacey et al.*, 1960]; shape parameter $T = (2\eta_2-\eta_1-\eta_3)/(\eta_1-\eta_3)$ [*Jelinek*, 1981; *Hrouda*, 1982]; and corrected anisotropy degree $P' = \exp\{2[(\eta_1-\eta_m)^2 + (\eta_2-\eta_m)^2 + (\eta_3-\eta_m)^2]\}^{-1/2}$, where $\eta_1 = \ln K_{max}$, $\eta_2 = \ln K_{int}$, $\eta_3 = \ln K_{min}$, and $\eta_m = (\eta_1+\eta_2+\eta_3)/3$ and $K_1=K_{max}$, $K_2=K_{int}$, $K_3=K_{min}$ [*Jelinek*, 1981].

In Figure 10 we show the directions of principal axes of the AMS ellipsoids for all of the specimens from each of the three Erap Valley thrust units. In Figure 11 we plot the $T$ and $P'$ parameters that describe the shape of the ellipsoid and its degree of anisotropy. Finally, in Table 2 we show the results obtained by tensor averaging the data for each thrust unit using the Hext/Jelinek method [*Hext*, 1963; *Jelinek*, 1978] as formulated by *Lienert* [1991]. Average bulk susceptibility decreases progressively from upper to lower Erap Valley thrusts (from about $6.0 \times 10^{-2}$ to $4.3 \times 10^{-2}$ SI units), perhaps reflecting a decrease in volcaniclastic input through time.

The upper projections in Figure 10 show principal AMS axes in stratigraphic (bedding-tilt corrected) coordinates, and the lower projections show these directions after the vertical axis rotations indicated by the mean ChRM declinations for each thrust sheet have been removed. The AMS principal axes reported here yield oblate ellipsoids with foliation near the bedding plane and a subordinate lineation shown by the maximum axis ($K_{max}$). The minimum axis ($K_{min}$) forms a partial girdle in the vertical plane perpendicular to the $K_{max}$ axis (Figure 10). The foliated shape is characteristic of primary sedimentary fabrics, with or without post depositional compaction [*Tarling and Hrouda*, 1993].

The key question for our purposes is the origin of the magnetic lineation. Magnetic lineations can be characteristic of slope fabrics acquired during fluvial deposition. In the Erap Valley data, however, streaking of $K_{min}$ perpendicular to $K_{max}$ contradicts a depositional current or slope fabric, in which $K_{min}$ tends to deviate from vertical as a cluster [*Taira*, 1989]. Secondary magnetic lineations may also be imparted through transitional tectonic strains as a "pencil structure" [*Graham*, 1978; *Kligfield et al.*, 1981]. However, the Erap

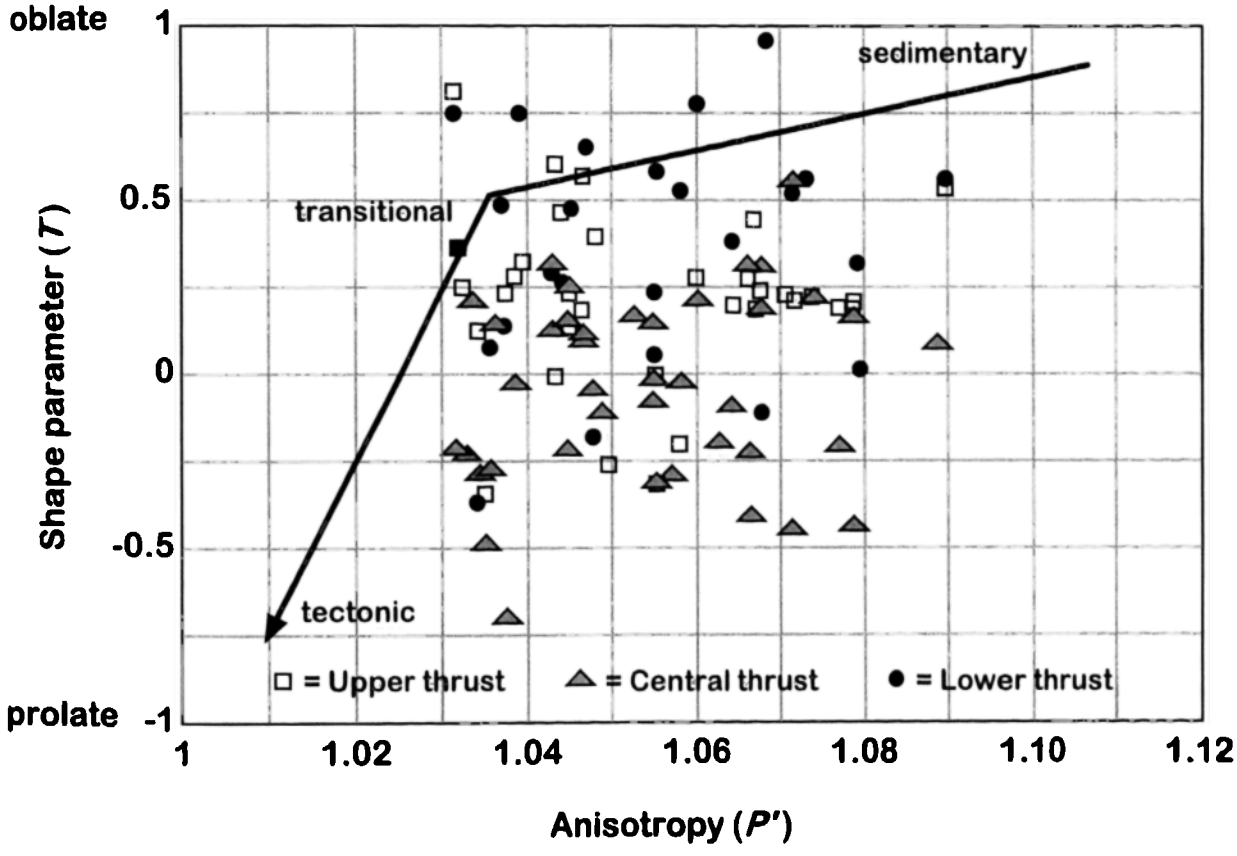



**Figure 11.** AMS fabric diagram for Erap Valley thrust units. $T$ is the shape parameter and $P'$ is the corrected degree of anisotropy (see text) for each specimen. Bold line indicates general path of typical sedimentary fabrics during strain (adapted from Kanamatsu et al. [1996]).

19449194, 1997, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/97TC00493 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**Table 2.** AMS Site Statistics (Restored)

| | Mean D, deg | Mean I, deg * | $\alpha_{95}$, deg † | Mean $K$ ($10^{-3}$ SI) | Mean $P'$ | Mean $T$ ‡ |
|---|---|---|---|---|---|---|
| | | | *Upper Erap Thrust* | | | |
| | | | | 59.7 +/-25.9 | 1.055 | 0.235 |
| $K_{max}$: | 290.3 | -2.6 | 9:5 | | | |
| $K_{int}$: | 19.8 | 12.0 | 8:7 | | | |
| $K_{min}$: | 212.2 | 77.7 | 8:5 | | | |
| $K_{max}/K_{int}$ = 1.017 | | $K_{int}/K_{min}$ = 1.026 | $K_{max}/K_{min}$ = 1.044 | | | |
| | | | *Central Erap Thrust* | | | |
| | | | | 53.1 +/-25.8 | 1.050 | -0.031 |
| $K_{max}$: | 297.6 | -7.0 | 10:5 | | | |
| $K_{int}$: | 26.3 | 10.5 | 10:5 | | | |
| $K_{min}$: | 240.8 | 77.4 | 7:4 | | | |
| $K_{max}/K_{int}$ = 1.016 | | $K_{int}/K_{min}$ = 1.021 | $K_{max}/K_{min}$ = 1.037 | | | |
| | | | *Lower Erap Thrust* | | | |
| | | | | 43.3 +/-21.6 | 1.018 | 0.362 |
| $K_{max}$: | 283.3 | 7.4 | 17:6 | | | |
| $K_{int}$: | 15.3 | 14.6 | 17:13 | | | |
| $K_{min}$: | 167.1 | 73.5 | 15:5 | | | |
| $K_{max}/K_{int}$ = 1.003 | | $K_{int}/K_{min}$ = 1.008 | $K_{max}/K_{min}$ = 1.011 | | | |

* Principal axes reported are restored through the vertical axis rotations given by the mean charactreristic remanent magnetization declination for each thrust sheet.

† Confidence ellipses are from Hext/Jelinek statistical analysis (see text).

‡ $P'$ is the "corrected anisotropy degree" of *Jelinek* [1981], and $T$ is the "shape parameter" of *Hrouda* [1982] (see text).

Valley thrusts are only weakly strained, with no macroscopic evidence of strain deformation except in the immediate fault zones and thus do not exhibit a true "pencil structure."

Since the thrust faults in Erap Valley are nearly parallel to bedding, we can explain the development of the lineation as the result of a strain-induced rotation of magnetic grains away from the foliation plane about an axis perpendicular to tectonic transport (the Y direction). This can have the effect of abruptly reorienting the $K_{max}$ direction parallel to Y (F. Hrouda, personal communication, 1996). For example, consider an assemblage of magnetic grains with an initial typical sedimentary fabric that has $K_{min}$ perpendicular to bedding (in the Z direction) and $K_{int}$ and $K_{max}$ of nearly equal magnitude with $K_{max}$ oriented downstream in approximately the X direction (i.e., away from the mountain range). Any shortening parallel to the bedding plane in a direction X will tilt grains such that their individual $K_{max}$ axes rotate away from the X-Y plane about the Y axis thereby weakening $K_{max}$ of the bulk fabric but leaving $K_{int}$ unchanged. At some point, $K_{max}$ and $K_{int}$ of the fabric will become equal, and with the next increment of strain their orientations will exchange so that $K_{max}$ is parallel to Y. The end result, after very little strain, will be a significant grouping of the $K_{max}$ axes of samples in the thrust sheet perpendicular to the direction of tectonic transport, just as observed in Figure 10.

Our fabrics are similar to those characterized in a study of magnetic fabrics from the accretionary complex of the Boso and Miura Peninsulas of Japan [*Kanamatsu et al.*, 1996]. In that study, sedimentary magnetic fabric shapes were found to follow a path of decreasing shape parameter ($T$) and decreasing corrected anisotropy ($P'$) in an evolution from sedimentary through transitional to "tectonic" fabrics. Data from the Leron Formation thrust sheets cluster in the domain interpreted by *Kanamatsu et. al* [1996] as early superposition of a tectonic strain deformation upon a primary, sedimentary fabric (Figure 11).

As shown in the lower projections of Figure 10, AMS principal axes from the three Erap Valley thrust sheets have very similar orientations when the vertical-axis rotations indicated by the ChRM declination anomalies are reverse rotated. Agreement of $K_{max}$ axes is within 15° when the three ellipsoids are restored by clockwise rotation through the vertical axis rotations indicated by our interpretation of ChRM polarity. The mean $K_{max}$ goes to 290.3° for the upper, 297.6° for the central, and 283.3° for the lower units, respectively (Figure 10). This agreement of the restored AMS principal

19449194, 1997, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/97TC00493 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**Table 3.** ChRM Directions

| Specimen | G. Dec | G. Inc | S. Dec | S. Inc | Strike/Dip* |
|---|---|---|---|---|---|
| | | *Lower Erap Thrust* | | | |
| 1c | 339.8 | -6.5 | 346.1 | -30.4 | 215/30 |
| 2c | 325.8 | -11.7 | 326.0 | -36.7 | 235/25 |
| 3c | 331.5 | -8.8 | 329.3 | -27.7 | 260/20 |
| 4c | 331.8 | -7.1 | 329.8 | -26.0 | 260/20 |
| 5b | 355.7 | -207 | 334.6 | -22.0 | 260/20 |
| 6b | 335.0 | -7.8 | 333.3 | -27.1 | 260/20 |
| 7c | 328.3 | -10.6 | 325.4 | -29.0 | 260/20 |
| 8d | 355.8 | -15.3 | 356.8 | -35.2 | 260/20 |
| 9c | 333.3 | 1.9 | 332.5 | -17.2 | 260/20 |
| 10b | 332.2 | 6.9 | 331.9 | -12.1 | 260/20 |
| 11b | 334.0 | 1.3 | 333.2 | -17.9 | 260/20 |
| 12c | 327.7 | 18.5 | 327.7 | -18.3 | 260/20 |
| 13b | 349.5 | 23.1 | 349.8 | -19.3 | 270/43 |
| 14c | 357.3 | 19.9 | 356.3 | -28.9 | 280/50 |
| 15d | 344.6 | 16.0 | 341.8 | -29.2 | 280/50 |
| 16b | 344.4 | 14.6 | 341.0 | -30.4 | 280/50 |
| 17b | 350.7 | 8.1 | 345.2 | -38.7 | 280/50 |
| 18b | 352.6 | 1.1 | 348.6 | -40.7 | 275/43 |
| 19b | 341.8 | -11.4 | 328.6 | -49.4 | 275/43 |
| 20c | 344.5 | -1.5 | 337.3 | -41.1 | 275/43 |
| 21b | 327.6 | 1.0 | 319.3 | -31.9 | 275/43 |
| 22b | 352.7 | -5.8 | 346.7 | -47.5 | 275/43 |
| 23b | 0.2 | -9.2 | 357.3 | -52.0 | 275/43 |
| 24b | 349.0 | -8.8 | 340.2 | -49.4 | 275/43 |
| 25b | 346.0 | 23.7 | 346.8 | -17.2 | 275/43 |
| 26b | 344.9 | 12.3 | 342.6 | -28.0 | 275/43 |
| | | *Central Erap Thrust* | | | |
| 27b | 324 | 11.2 | 322.7 | -42.6 | 238/54 |
| 28b | 89.9 | -26.5 | 98.1 | 6.9 | 238/54 |
| 29b | 95.6 | -24.5 | 100.6 | 11.9 | 238/54 |
| 30b | 89.7 | -32.9 | 102.4 | 2.2 | 238/54 |
| 31a | 102.4 | -12.1 | 92.4 | 37.1 | 228/62 |
| 32a | 106.3 | -11.1 | 95.5 | 40.3 | 228/62 |
| 33a | 103.9 | -13.9 | 95.3 | 36.7 | 228/62 |
| 35b | 67.8 | -33.6 | 84.8 | -1.1 | 225/58 |
| 36b | 62.3 | -36.3 | 84.3 | -6.3 | 225/58 |
| 37b | 47.7 | -33.6 | 75.5 | -15.1 | 225/58 |
| 38b | 81.8 | -29.8 | 88.5 | 12.4 | 220/57 |
| 39b | 101.3 | -29.2 | 103.1 | 22.1 | 220/57 |
| 40b | 89.3 | -34.7 | 96.7 | 12.3 | 220/57 |
| 41b | 89.8 | -33.7 | 95.3 | 14.7 | 215/56 |
| 42b | 87.1 | -29.3 | 90.9 | 17.3 | 215/56 |
| 43b | 63.9 | -47.9 | 90.4 | -8.3 | 218/58 |
| 44b | 55.8 | -46.0 | 85.5 | -11.6 | 218/58 |
| 45b | 54.7 | -37.4 | 77.9 | -7.4 | 218/58 |
| 46b | 51.8 | -15.1 | 56.9 | 6.9 | 217/66 |
| 47b | 54.8 | -32.2 | 71.6 | 7.4 | 215/74 |
| 48a | 53.9 | -39.4 | 77.9 | 3.8 | 215/74 |
| 49a | 59.6 | -34.4 | 75.4 | 10.0 | 215/74 |
| 50b | 60.8 | -26.4 | 69.7 | 11.2 | 215/68 |
| 51b | 61.8 | -17.3 | 62.1 | 16.7 | 215/68 |
| 52b | 44.8 | -34.5 | 70.4 | -4.7 | 215/68 |
| 53b | 74.8 | -35.6 | 86.7 | 13.0 | 220/69 |
| 54a | 84.9 | -32 0 | 89.7 | 21.7 | 220/69 |

19449194, 1997, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/97TC00493 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**Table 3.** (continued)

| Specimen | G. Dec | G. Inc | S. Dec | S. Inc | Strike/Dip* |
|---|---|---|---|---|---|
| | | *Central Erap Thrust* | | | |
| 55b | 73.7 | -45.1 | 93.8 | 6.4 | 220/69 |
| 56b | 49.6 | -36 | 79.9 | -10.4 | 225/66 |
| 57b | 37 | -43.5 | 84.3 | -21.9 | 225/66 |
| 58b | 60.2 | -43.5 | 90.3 | -6.1 | 225/66 |
| 59b | 35.9 | -57.2 | 98.9 | -24.8 | 225/66 |
| | | *Upper Erap Thrust* | | | |
| 60c | 116.0 | -53.6 | 124.8 | 14.0 | 228/70 |
| 61b | 116.5 | -47.5 | 122.7 | 19.8 | 228/70 |
| 62b | 80.2 | -49.4 | 103.4 | 3.1 | 226/68 |
| 63c | 86.8 | -39.8 | 99.3 | 13.0 | 226/68 |
| 65b | 100.9 | -51.3 | 114.5 | 11.4 | 226/68 |
| 66b | 99.6 | -55.0 | 115.9 | 7.9 | 226/68 |
| 67b | 96.5 | -44.8 | 104.7 | 12.7 | 215/62 |
| 69b | 89.7 | -43.9 | 99.9 | 11.2 | 215/62 |
| 70b | 68.7 | -52.2 | 94.3 | -4.1 | 215/62 |
| 71b | 62.5 | -65.8 | 102.9 | -15.1 | 215/62 |
| 73a | 88.2 | -66.6 | 111.1 | -8.6 | 215/62 |
| 74b | 125.5 | -53.7 | 129.3 | 9.9 | 225/64 |
| 75a | 133.4 | -63.2 | 134.3 | 0.8 | 225/64 |
| 77b | 124.2 | -63.9 | 124.6 | 4.1 | 215/68 |
| 79b | 94.9 | -82.0 | 120.9 | -15 | 215/68 |
| 81b | 93.9 | -74.4 | 124.4 | -16.8 | 225/61 |
| 83b | 107.5 | -54.7 | 119.5 | 3.0 | 225/61 |
| 86b | 134.0 | -63.7 | 137.3 | -0.8 | 230/63 |
| 87b | 138.2 | -55.5 | 137.7 | 4.5 | 227/60 |
| 89b | 105.4 | -43.6 | 114.3 | 10.9 | 227/60 |
| 91b | 117.8 | -46.9 | 127.0 | 7.1 | 236/58 |
| 92b | 121.9 | -43.5 | 128.4 | 11.3 | 236/58 |
| 93b | 99.5 | -50.7 | 118.6 | -2.3 | 236/58 |
| 95b | 109.2 | -38.3 | 117.3 | 11.8 | 236/58 |
| 97c | 120.9 | -29.7 | 121.5 | 26.7 | 233/60 |
| 99b | 91.8 | -31.5 | 100.3 | 11.6 | 233/60 |
| 100b | 59.9 | -45.4 | 97.9 | -19.7 | 236/58 |
| 103b | 94.1 | -44.9 | 112.1 | -0.2 | 236/58 |
| 105c | 52.8 | -42.7 | 93.0 | -23.2 | 236/58 |
| 108c | 142.2 | -61.9 | 137.9 | 1.9 | 224/64 |
| 110b | 142.6 | -33.6 | 142.3 | 29.8 | 224/64 |
| 111b | 116.9 | -28.6 | 118.0 | 25.6 | 244/65 |
| 112b | 119.6 | -29.2 | 120.6 | 26.5 | 244/65 |
| 113b | 95.2 | -60.9 | 123.8 | -2.4 | 236/68 |
| 115b | 125.8 | -38.2 | 128.3 | 26.9 | 236/68 |
| 119c | 77.7 | -42.8 | 101.6 | -4.8 | 234/60 |
| 120b | 108.6 | -46.4 | 116.9 | 10.0 | 225/60 |
| 123b | 110.5 | -51.3 | 119.9 | 5.9 | 225/60 |
| 124b | 120.7 | -45.1 | 126.4 | 13.6 | 231/61 |

G. Dec, geographic (in situ) declination; G. Inc, geographic (in situ) inclination; S. Dec, stratigraphic (tilt-corrected) declination; S. Inc, stratigraphic (tilt-corrected) inclination.

* Strike is recorded to left of downdip direction.

19449194, 1997, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/97TC00493 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

axes directions provides independent confirmation of the vertical axis rotations and indicates that the simple bedding-tilt correction applied to these data is well suited for interpretation of these thrust sheets.

The coincidence of principal axes after restoration of the vertical-axis rotations provides additional evidence of a tectonic origin for the lineation, since such consistent directions would seem unlikely to be sedimentary fabrics acquired in a braided fan depositional environment. $K_{min}$ axes in all three data sets partially girdle the lineation, suggesting a strain-induced smearing of initially well-clustered minima.

### Kinematics and Rotation Rate

Data from this study document large, counterclockwise, vertical-axis rotations of thrust sheets of the Leron Formation exposed in the Erap Valley. A plausible interpretation of these rotations is that they are kinematically related to shear along the northern extension of the sinistral Wampit Fault (Figure 2). This would imply that the Erap Valley follows the expression of a tear-fault, which acts at depth to transfer slip parallel to the thrust direction. This tear fault likely inherits its location and sense of motion from the Wampit Fault south of the RMFZ, and the rotations of the Leron Formation in Erap Valley are most simply interpreted as the result of shear due to strike-slip displacement on such a structure.

Mean ChRM directions from the upper thrust sheet imply that a unit over 500 m thick and at least 1 km in length was rotated counterclockwise by 60° in less than 1 m.y. This rate of vertical-axis rotation is much higher than the highest minimum Neogene rotation rates including 6.3°/m.y. and 5°-10°/m.y. in the Transverse Ranges of California [*Jackson and Molnar*, 1990; *Luyendyk*, 1991] and 10°/m.y. in the coastal range of eastern Taiwan [*Lee et al.*, 1991]. However, the plan view extent of the rotated units is small and difficult to assess exactly because fresh outcrops are restricted to river exposures. Large declination anomalies were not observed in reconnaissance samples from the Leron Formation 7 km to the southeast at Camp Diddy which provides a southeastern limit to the rotated units.

These rapid rotations are also intriguing because they vary widely between thrust sheets in an imbricate thrust stack, with abrupt changes in the amount of vertical-axis rotation at fault boundaries (Table 3). Declination anomalies do not increase monotonically with age through the sampled section but, rather, are structurally controlled. The observation that declination anomalies are quite distinct in the three thrust sheets argues against postthrusting rotation, and the discontinuities in declination anomalies at the thrust boundaries imply that rotation was not gradual during deposition but instead occurred during thrust sheet emplacement. Thus it is likely that the rotations occurred much more rapidly than the minimum rate of 60° to 90°/m.y. Larger-scale contemporaneous thrusting and rotation are reported by *Channell et al.* [1990] and *Oldow et al.* [1990] in the western Sicilian thrust belt. Although the Sicilian rotations are associated with a major change in tectonic transport direction through time rather than with local shear across a high-angle fault, contemporaneous thrusting and rapid vertical-axis rotation may be a more common process in fold and thrust belts than is presently recognized.

## Conclusions

Significant findings of this study are (1) the observation of large and discontinuous declination anomalies which are related to tectonic transport during thrust sheet emplacement, (2) very rapid vertical-axis rotations, (3) information pertaining to the development of early magnetic fabrics in weakly strained sedimentary thrust units, and (4) a constraint on the age of the Leron Formation in the Erap Valley area.

On the basis of demagnetization behavior, magnetic fabrics, and geologic evidence, we interpret the upper and central thrusts in Erap Valley to be of Matuyama (reversed polarity) age (>780 ka). This interpretation is supported by the reversely magnetized Leron Formation at Camp Diddy to the SE. The lower Erap thrust sheet exhibits a syn tilting remagnetization of normal polarity (<780 ka). We infer that the rapid counterclockwise rotations of units exposed in Erap River are restricted to the Erap Valley area and are kinematically related to sinistral displacement on the Wampit Fault, which continues as a young structure across the Ramu-Markham Valley.

The primary significance of this study is the observation that these rapid vertical-axis rotations were contemporaneous with thrusting, rather than acquired gradually during deposition. The implication is that caution must be exercised in interpretation of paleomagnetic data in fold and thrust belts, since small blocks may be chaotically rotated, and these rotations may be very rapid and poorly estimated by time-averaged minimum rates.

**Acknowledgments.** We gratefully acknowledge the financial support for this work provided by the National Science Foundation (NSF grant EAR-9406652). Financial assistance was also provided by a Geological Society of America H.T. Stearns Fellowship award. Bob Butler, Hugh Davies, Bob Findlay, Frantisek Hrouda, Bruce Luyendyk, and Eli Silver provided helpful discussions and reviews, and Barry Lienert kindly gave us a copy of his APLOT11 software. Finally, this study would not have been possible without the unselfish cooperation and unending generosity and guidance of Lon Abbott.

## References

Abbott, L.D., Structural and sedimentological history of a progressive arc-continent collision: The Finisterre Range, northern Papua New Guinea, Ph.D. thesis, Univ. of Calif., Santa Cruz, 1993.

Abbott, L.D, and E A Silver, Geology of the southern Finisterre Range. A case history of modern arc-continent collision, in *Proceedings of the PNG Geology Exploration and Mining Conference 1991, Rabaul*, edited by R. Rogerson, pp. 1-7, Australas. Inst. of Min. and Metall., Melbourne, 1991.

Abbott, L D, E.A. Silver and J Galewsky, Structural evolution of a modern arc-continent collision in Papua New Guinea, *Tectonics*, *13*(5), 1007-1034, 1994.

Abers, G.A., and R. McCaffrey, Active arc-continent collision: Earthquakes, gravity anomalies, and fault kinematics in the Huon-Finisterre collision zone, Papua New Guinea, *Tectonics*, *13*(2), 227-245, 1994.

19449194, 1997, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/97TC00493 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Balsey, J.R., and A.F. Buddington, Magnetic susceptibility anisotropy and fabric of some Adirondack granites and orthogneisses, *Am. J. Sci., 2,* 6-20, 1960.

Beck, M.E., Discordant paleomagnetic pole position as evidence of regional shear in the western Cordillera of North America, *Am. J. Sci., 276,* 694-712, 1976.

Brierley G.J., K. Liu, and A.W. Crook, Sedimentology of coarse-grained alluvial fans in the Markham Valley, Papua New Guinea, *Sediment. Geol., 86,* 297-324, 1993.

Cande, S.C., and D.V. Kent, A new geomagnetic polarity time scale for the Late Cretaceous and Cenozoic, *J. Geophys. Res., 97,* 13917-13951, 1992.

Channell, J.E.T., J.S. Oldow, R. Catalano, and B. D'Argenio, Paleomagnetically determined rotations in the western Sicilian fold and thrust belt, *Tectonics, 9*(4), 641-660, 1990.

Coe, R.S., B.R. Globerman, P.W. Plumley, and G.A.Thrupp, Paleomagnetic results from Alaska and their tectonic implications, in *Tectonostratigraphic Terranes of the Circum-Pacific Region, Circum-Pac. Counc. Energy Miner. Resour. Ser., vol. 1,* edited by D.G. Howell, pp.85-108, Am. Assoc. Pet. Geol., Houston, Tex., 1985.

Fuller, M., R. Haston, and E. Schmidtke, Paleomagnetism in SE Asia: sinistral shear between Philippine Sea plate and Asia, in *Paleomagnetic Rotations and Continental Deformation, Proceedings of the NATO Advanced Research Workshop on Paleomagnetic Rotations and Continental Deformation, Loutra Edipsou, Greece,* edited by C. Kissel and C. Laj, pp. 411-430, Kluwer, Dordrecht, The Netherlands, 1989.

Graham, R.H., Quantitative deformation studies in the Permian rocks of the Alpes Maritimes, Proceedings of the symposium in honour of Prof. J. Goguel, *Mem. Bur. Rech. Geol. Min., 91,* 219-238, 1978.

Hext, G.R., The estimation of second order tensors, with related texts and designs, *Biometrika, 50,* 353-373, 1963.

Hrouda, F., Magnetic anisotropy of rocks and its application in geology and geophysics, *Geophys. Surv., 5,* 37-82, 1982.

Hrouda, F., Models of magnetic anisotropy variations in sedimentary thrust sheets, *Tectonophysics, 185,* 203-210, 1991.

Jackson, J., and P. Molnar, Active faulting and block rotations in the western transverse Ranges, *J. Geophys. Res., 95,* 22073-22087, 1990.

Jelinek, V., Statistical processing of anisotropy of magnetic susceptibility measured on groups of specimens, *Stud. Geophys. Geod., 22,* 50-62, 1978.

Jelinek, V., Characterization of the magnetic fabric of rocks, *Tectonophysics, 79,* 63-67, 1981.

Kanamatsu, T., E. Herrero-Bervera, A. Taira, J. Ashi, and A.S. Furumoto, Magnetic fabric development in the Tertiary accretionary complex in the Boso and Miura peninsulas of central Japan, *Geophys. Res. Lett., 23,* 471-474, 1996.

Kligfield, R., W.H. Owens, and W. Lowrie, Magnetic susceptibility anisotropy, strain, and progressive deformation in Permian sediments from the Maritime Alps (France), *Earth Planet. Sci. Lett., 55,* 181-189, 1981.

Lee, T.Q., C. Kissel, E. Barrier, C. Laj, and C. Wen-Rong, Paleomagnetic evidence for a diachronic clockwise rotation of the coastal range, eastern Taiwan, *Earth Planet. Sci. Lett., 104,* 245-257, 1991.

Lienert, B.R., Monte Carlo simulations of errors in the anisotropy of magnetic susceptibility: A second-rank symmetric tensor, *J. Geophys. Res., 96,* 19539-19544, 1991.

Liu, K., Sedimentation and tectonics of the Markham suture, Papua New Guinea, Ph.D. thesis, Aust. Nat. Univ., Canberra, Aust. Capital Terr., 1993.

Liu, K., and A.W. Crook, Miocene-Pliocene deep sea to alluvial fan delta sedimentation in the Markham Basin, Papua New Guinea: Sedimentary response to arc-continent collision, in *Proceedings of the Second PNG Petroleum Convention, Port Moresby,* edited by G.J. Carman and Z. Carman, pp. 97-109, Australas. Inst. of Min. and Metall., Melbourne, 1993.

Luyendyk, B.P., A model for Neogene crustal rotations, transtension, and transpression in southern California, *Geol. Soc. Am. Bull., 103,* 1528-1536, 1991.

Luyendyk, B.P., M.J. Kammerling, R.R.Terres, and J.S.Hornafius, Simple shear of southern California during Neogene time suggested by paleomagnetic declinations, *J. Geophys Res., 90,* 12454-12466, 1985.

McFadden, P.L., A new fold test for palaeomagnetic studies, *Geophys. J. Int., 103*(1), 163-169, 1990.

McFadden, P.L., and M.W., McElhinny, The combined analysis of remagnetization circles and direct observations in palaeomagnetism, *Earth Planet. Sci. Lett., 87,* 161-172, 1988

Oldow, J.S., J.E.T. Channell, R. Catalano, and B D'Argenio, Contemporaneous thrusting and large-scale rotations in the western Sicilian fold and thrust belt, *Tectonics, 9*(4), 661-681, 1990

Pigram, C.J., and H.L. Davies, Terranes and the accretion history of the New Guinea Orogen, *BMR J. Aust. Geol. Geophysics, 10,* 193-211, 1987.

Schwartz, S.Y., and R. Van der Voo, Thrust sheet rotation during foreland impingement in the Wyoming-Idaho overthrust belt, *J. Geophys Res., 89,* 10077-10086, 1984.

Silver, E.A., L.D. Abbott, K.S. Kirchoff-Stein, D.L. Reed, B. Bernstein-Taylor, and D Hilyard, Collision propagation in Papua New Guinea and the Solomon Sea, *Tectonics, 10*(5), 863-874, 1991.

Simpson, R.W., and A. Cox, Paleomagnetic evidence for tectonic rotation of the Oregon Coast Range, *Geology, 5,* 585-589, 1977.

Stacey, F.D., G. Joplin, and J. Lindsay, Magnetic anisotropy and fabric of some foliated rocks from S.E. Australia, *Geofi. Pura Appl., 47,* 30-40, 1960.

Taira, A., Magnetic fabrics and depositional processes, in *Sedimentary Facies in the Active Plate Margins,* edited by A. Taira and F Masuda, pp. 43-77, Terra Sci., Tokyo, 1989

Tarling, D.H., and F. Hrouda, The Magnetic Anisotropy of Rocks, 217 pp., Chapman and Hall, New York, 1993.

Tingey, R.J., and D.J. Grainger, Markham, Papua New Guinea, geological series map and explanatory notes, 1:250,000. Geol. Surv of Papua New Guinea, Port Moresby, 1976

Wells, R.E., and P.L. Heller, The relative contribution of accretion, shear, and extension to Cenozoic tectonic rotation in the Pacific Northwest, *Geol. Soc. Am. Bull., 100,* 325-338, 1988.

R. S. Coe and P. D. Weiler, Department of Earth Sciences, University of California, Santa Cruz, Earth and Marine Sciences Building, Santa Cruz, CA 95064. (e-mail: rcoe@earthsci.ucsc edu, pweiler@earthsci.ucsc.edu

(Received July 31, 1996;
revised December 26, 1996;
accepted January 11, 1997.)

19449194, 1997, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/97TC00493 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License